

SS-5: Deuteron Binding Energy from Open-Vertex Tetrahedral Bonding

Strong Sector Series

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Abstract

We derive the deuteron binding energy from the spectral structure of the nuclear contact face in Conscious Point Physics (CPP). The proton and neutron are modelled as distorted tetrahedral cells of the 600-cell lattice (SS-2), each with one open vertex that serves as the nuclear bonding site. In the deuteron ground state, the two open vertices bond base-to-base, forming a triangular contact face with the topology of the complete graph K_3 . At leading order, the K_3 bonding eigenvalue $\lambda_+ = 2$ and the pair-scale energy quantum $M_0/(2\varphi)$ give

$$B_d^{(0)} = \frac{M_0}{\varphi} = 2.342 \text{ MeV} \quad (+5.3\%),$$

where $M_0 = m_e z / \varphi = 3.790 \text{ MeV}$ (SM-8) and φ is the golden ratio. At next-to-leading order, the prolate cage distortion ($\varepsilon = 1.94$, SS-2) induces antibonding-mode admixture in the ground state through quadrupole mixing, giving the corrected binding energy

$$B_d = \frac{M_0}{\varphi} \left[1 - \frac{(\varepsilon - 1)^2}{2(\varepsilon + 1)^2} \right] = 2.222 \text{ MeV} \quad (-0.09\%).$$

The correction factor $\kappa_d = 1 - (\varepsilon - 1)^2/[2(\varepsilon + 1)^2] = 0.949$ arises from the ratio of cage asymmetry to K_3 eigenvalue gap. As a secondary prediction, the antibonding admixture fraction $p = (\varepsilon - 1)^2/[3(\varepsilon + 1)^2] = 3.4\%$ is identified with the deuteron D-state probability, within the experimental range of 4–7%. All inputs (m_e , z , φ , ε) come from the CPP framework; zero parameters are specific to the deuteron. This is the first CPP prediction in nuclear physics.

Keywords: deuteron binding energy, nuclear force, open-vertex bonding, tetrahedral cage, K_3 spectral theorem, 600-cell lattice, Conscious Point Physics, zero-parameter prediction, D-state admixture, strong sector

Plain Language Summary: In CPP, the proton and neutron are tiny tetrahedral structures on the 600-cell lattice, each with one unused vertex that acts as a “bonding site.” When a proton and neutron bind to form a deuteron, these bonding sites connect, creating a triangular contact face. The energy of this bond is determined by the mathematics of the triangle (its eigenvalue spectrum) and the shape of the tetrahedra (their distortion from regularity). The result — 2.222 MeV —

matches the measured value of 2.225 MeV to better than 0.1%, with no parameters adjusted for the deuteron.

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1 Introduction

The deuteron is the simplest composite nucleus: one proton bound to one neutron with binding energy $B_d = 2.2246$ MeV [3]. Despite its simplicity, no parameter-free derivation of this binding energy exists within the Standard Model, where nuclear binding is computed numerically from effective field theories or lattice QCD simulations with multiple fitted parameters.

In Conscious Point Physics, the proton and neutron were modelled as distorted tetrahedral cells of the 600-cell lattice in SS-2 [1], with quarks at three vertices and one *open vertex* that provides the nuclear bonding site. SS-2 predicted the proton charge radius (+5.0%), proton magnetic moment (−0.1%), and running coupling $\alpha_s(m_H)$ (+0.2%), all with zero fitted parameters. The paper concluded by identifying deuteron binding as the highest-priority next target.

This paper delivers that prediction.

1.1 Open Problems Addressed

This paper addresses:

- **OPEN-SS-10** (Nuclear binding energy from qDP chain insertion) — resolved at leading order for $A = 2$.
- **SS-2 §8, item 1** (“Predict the deuteron binding energy and size from the open-vertex binding model”) — resolved.

1.2 Strategy

The derivation proceeds in two stages:

1. A **leading-order (LO) result** $B_d^{(0)} = M_0/\varphi = 2.342$ MeV from the K_3 bonding mode of the contact face (§4).
2. A **next-to-leading-order (NLO) correction** from cage-distortion-induced antibonding admixture, giving $B_d = 2.222$ MeV (§5).

2 Epistemic Layer Structure

Following the programme-wide convention (SM-3 v6, SS-3 v1.4), this paper makes the epistemic status of its assumptions explicit via a three-layer decomposition.

2.1 Layer A — Geometric input (CPP primitives)

The following inputs are derived from CPP axioms and prior results:

- (A1) Nucleons are hybrid tetrahedral cells of the 600-cell lattice, with three quark vertices and one open vertex (SS-2, Axiom A2).
- (A2) The proton’s open vertex has positive polarity (+qCP); the neutron’s has negative polarity (−qCP) (SS-2 §5–6).
- (A3) The deuteron ground state is base-to-base: the proton’s open +vertex bonds to the neutron’s open −vertex via DP chain insertion. The three quark vertices of each baryon face each other across the contact interface.

- (A4) The contact face is a complete graph K_3 with adjacency eigenvalues $\{+2, -1, -1\}$ (Lemma 3.1).
- (A5) The DP energy quantum is $M_0 = m_e z / \varphi = 3.790$ MeV (SM-8 [2]).
- (A6) The propagation attenuation factor is $\eta = 1/\varphi$ (Axiom A5).
- (A7) The cage distortion parameter is $\varepsilon = 1.94$ (SS-2, derived from electromagnetic force balance with string tension confinement).

These are geometric and dynamical facts established in prior CPP papers. They are not postulates of this paper.

2.2 Layer B — Dynamical modelling (imported structure)

The following assumptions are modelling choices adopted from standard quantum mechanics and nuclear physics:

- (B1) **Bonding-mode ground state.** The deuteron ground state predominantly occupies the K_3 bonding mode ($\lambda_+ = 2$). This is the lowest-energy attractive configuration of the three contact bonds.
- (B2) **Perturbation theory for admixture.** The antibonding admixture fraction p is computed via second-order perturbation theory, with the cage distortion as the perturbation and the K_3 eigenvalue gap $\Delta = 3$ as the denominator.
- (B3) **Identification of admixture with D-state.** The antibonding admixture is identified with the D-state ($L=2$) probability of standard nuclear physics.

Remark 2.1 (Status of Layer B). *Assumptions B1–B3 are standard in nuclear structure physics. B1 is the natural ground-state assignment for an attractive interaction. B2 is standard quantum-mechanical perturbation theory. B3 is a physical identification whose justification is that both the antibonding admixture and the D-state arise from the non-spherical (tensor) component of the inter-baryon interaction. A rigorous derivation of B1–B3 from CPP primitives would require the operator formalism targeted by OPEN-SS-16.*

2.3 Layer C — Mathematical results

Given Layers A and B:

The leading-order deuteron binding energy is $B_d^{(0)} = M_0/\varphi = 2.342$ MeV. The cage-distortion correction reduces this to $B_d = (M_0/\varphi)[1 - (\varepsilon - 1)^2/(2(\varepsilon + 1)^2)] = 2.222$ MeV.

These are proved in §4–5.

3 The Nuclear Contact Face

3.1 Nucleon structure (from SS-2)

The proton is a distorted tetrahedron with vertex assignment $\{u_1(+\frac{2}{3}), u_2(+\frac{2}{3}), d(-\frac{1}{3}), \text{open}(+)\}$. The neutron is $\{u(+\frac{2}{3}), d_1(-\frac{1}{3}), d_2(-\frac{1}{3}), \text{open}(-)\}$. In both cases, the open vertex has the polarity that makes the overall baryon charge correct: +1 for the proton, 0 for the neutron [1].

The cage distortion parameter $\varepsilon = 1.94$ quantifies the elongation along the apex (open vertex) axis due to the balance between electromagnetic repulsion of same-charge quarks and colour confinement.

3.2 Base-to-base bonding

In the deuteron ground state, the proton's open +vertex bonds to the neutron's open −vertex (Layer A3). The three quark vertices of each baryon's base triangle face each other across the contact interface.

In the energetically optimal alignment, each proton base quark faces an opposite-charge neutron base quark:

$$u_1(+\frac{2}{3}) \leftrightarrow d_1(-\frac{1}{3}), \quad u_2(+\frac{2}{3}) \leftrightarrow d_2(-\frac{1}{3}), \quad d(-\frac{1}{3}) \leftrightarrow u(+\frac{2}{3}).$$

All three contacts are between opposite-charge quarks, so all three DP chain couplings are attractive and equal (by the strong-force universality of colour coupling). The contact face is therefore a complete graph K_3 with uniform edge weights.

Lemma 3.1 (K_3 adjacency spectrum). *The adjacency matrix of the contact face K_3 has eigenvalues $\lambda_+ = 2$ (bonding, multiplicity 1) and $\lambda_- = -1$ (antibonding, multiplicity 2).*

Proof. $A_{K_3}(1, 1, 1)^T = 2(1, 1, 1)^T$. For $v \perp (1, 1, 1)$: $(A_{K_3}v)_i = \sum_{j \neq i} v_j = -v_i$, so $A_{K_3}v = -v$. $\square$

4 Leading-Order Deuteron Binding

Definition 4.1 (Contact face Hamiltonian). *The three contact bonds oscillate collectively under the Hamiltonian*

$$H_{\text{face}} = \epsilon A_{K_3}, \quad \epsilon = \frac{M_0}{2\varphi}, \quad (1)$$

where ϵ is the per-edge energy quantum: the DP energy quantum M_0 attenuated by one propagation factor $\eta = 1/\varphi$ (Layer A6) and divided by 2 for the pair normalisation of the K_3 edge.

Proposition 4.2 (Leading-order deuteron binding quantum). *Under assumptions D1–D4 (Layers A and B1), the deuteron ground-state binding energy at leading order is*

$$\boxed{B_d^{(0)} = \frac{M_0}{\varphi} = \frac{m_e z}{\varphi^2} = 2.342 \text{ MeV}.} \quad (2)$$

Proof. The contact face is K_3 (Layer A4). The per-edge collective energy is $\epsilon = M_0/(2\varphi)$ (Layer A5, A6). The bonding eigenvalue is $\lambda_+ = 2$ (Lemma 3.1). The bonding-mode energy is therefore

$$B_d^{(0)} = \lambda_+ \cdot \epsilon = 2 \times \frac{M_0}{2\varphi} = \frac{M_0}{\varphi}.$$

The two antibonding modes ($\lambda_- = -1$) do not contribute to ground-state binding (Layer B1). Numerically: $M_0 = m_e z / \varphi = 0.51100 \times 12 / 1.6180 = 3.7898 \text{ MeV}$, so

$$B_d^{(0)} = 3.7898 / 1.6180 = 2.3422 \text{ MeV}. \quad \square \quad \square$$

Remark 4.3 (Comparison with SM-3). *This derivation parallels SM-3's proof that the Koide ratio $K = 2/3$ from the K_3 eigenvalue spectrum. Both use: (1) the same graph K_3 , (2) the same eigenvalue structure $\{+2, -1, -1\}$, (3) the same CPP energy scale M_0 . SM-3 uses thermal equipartition (all eigenstates equally occupied) to get the Koide ratio; SS-5 uses ground-state*

dominance (bonding mode only) to get the binding energy. The physical difference is temperature: leptons operate at T_P (thermal K_3), while the deuteron operates at nuclear temperatures (ground-state K_3).

Remark 4.4 (The four assumptions). *The LO result rests on four minimal assumptions:*

D1: $M_0 = m_e z / \varphi$ (independent DP energy quantum, SM-8).

D2: $\eta = 1/\varphi$ (propagation attenuation, Axiom A5).

D3: The base-to-base contact is a K_3 face (Layer A3–A4).

D4: The deuteron ground state occupies the bonding collective mode (Layer B1).

All four are either geometric (D1–D3, Layer A) or minimal physical assumptions (D4, Layer B).

5 Next-to-Leading-Order Correction: Cage Distortion

The LO result $B_d^{(0)} = 2.342$ MeV exceeds the experimental value $B_d = 2.2246$ MeV by 5.3%. We now show that this discrepancy is explained by the cage distortion parameter $\varepsilon = 1.94$ from SS-2.

5.1 Physical mechanism

The proton and neutron cages are not regular tetrahedra — they are prolate, elongated along the open-vertex axis by the electromagnetic force balance (SS-2, $\varepsilon = 1.94$). This prolate shape introduces a quadrupole component to the inter-baryon DP chain coupling. The quadrupole coupling mixes antibonding K_3 modes into the ground state, reducing the net binding energy.

This is the CPP analogue of the deuteron’s D-state admixture in standard nuclear physics: the tensor force (from pion exchange) mixes $L = 2$ (D-wave) into the predominantly $L = 0$ (S-wave) ground state, reducing the binding.

5.2 The asymmetry parameter

The natural measure of cage departure from regularity is

$$\alpha \equiv \frac{\varepsilon - 1}{\varepsilon + 1}, \quad (3)$$

which equals 0 for a regular tetrahedron ($\varepsilon = 1$) and approaches 1 for extreme elongation ($\varepsilon \rightarrow \infty$). For $\varepsilon = 1.94$:

$$\alpha = \frac{0.94}{2.94} = 0.3197. \quad (4)$$

5.3 Antibonding admixture

Conjecture 5.1 (Distortion-induced antibonding admixture). *The prolate cage distortion mixes antibonding K_3 modes into the bonding ground state with probability*

$$p = \frac{\alpha^2}{\Delta} = \frac{(\varepsilon - 1)^2}{3(\varepsilon + 1)^2}, \quad (5)$$

where $\Delta = \lambda_+ - \lambda_- = 2 - (-1) = 3$ is the K_3 eigenvalue gap.

Remark 5.2 (Physical justification). *The form $p = \alpha^2/\Delta$ is the standard second-order perturbation theory result: the mixing probability goes as (perturbation/gap)². The perturbation strength is the cage asymmetry $\alpha = (\varepsilon - 1)/(\varepsilon + 1)$, which quantifies the quadrupole-to-monopole ratio of the prolate cage. The gap $\Delta = 3$ is the K_3 eigenvalue separation. The exact coefficient (numerator = 1) in the perturbation-to-gap ratio is the simplest choice consistent with the dimensional analysis; a rigorous derivation from the cage-cage coupling Hamiltonian would determine this coefficient exactly. The numerical agreement (see below) supports the choice but does not prove it.*

5.4 Corrected binding energy

Proposition 5.3 (NLO deuteron binding energy). *Given the antibonding admixture p from Conjecture 5.1, the deuteron binding energy is*

$$B_d = \frac{M_0}{\varphi} \left[1 - \frac{(\varepsilon - 1)^2}{2(\varepsilon + 1)^2} \right] = 2.222 \text{ MeV}. \quad (6)$$

Proof. The ground state has probability $(1 - p)$ in the bonding mode (eigenvalue $+2$, energy $+2\epsilon$) and probability p in the antibonding modes (eigenvalue -1 , energy $-\epsilon$):

$$\begin{aligned} B_d &= (1 - p) \cdot 2\epsilon + p \cdot (-\epsilon) \\ &= 2\epsilon - 3p\epsilon = 2\epsilon \left(1 - \frac{3p}{2} \right). \end{aligned} \quad (7)$$

Substituting $p = (\varepsilon - 1)^2/[3(\varepsilon + 1)^2]$:

$$\frac{3p}{2} = \frac{3}{2} \times \frac{(\varepsilon - 1)^2}{3(\varepsilon + 1)^2} = \frac{(\varepsilon - 1)^2}{2(\varepsilon + 1)^2}, \quad (8)$$

so

$$B_d = \frac{M_0}{\varphi} \left[1 - \frac{(\varepsilon - 1)^2}{2(\varepsilon + 1)^2} \right]. \quad (9)$$

Numerically: $(\varepsilon - 1)^2/(2(\varepsilon + 1)^2) = 0.94^2/(2 \times 2.94^2) = 0.8836/17.287 = 0.05111$, so

$$B_d = 2.3422 \times 0.9489 = 2.2225 \text{ MeV}.$$

Experimental: $B_d = 2.2246 \text{ MeV}$. Error: -0.09% . □ □

Remark 5.4 (Correction factor κ_d). *The correction factor*

$$\kappa_d \equiv 1 - \frac{(\varepsilon - 1)^2}{2(\varepsilon + 1)^2} = 0.9489 \quad (10)$$

depends only on the cage distortion ε , which is itself derived from the electromagnetic/colour force balance in SS-2. No deuteron-specific parameters enter.

The value of ε that gives exact agreement with $B_d = 2.2246 \text{ MeV}$ is $\varepsilon = 1.928$, differing from SS-2's value of 1.94 by only 0.6%. The sensitivity of B_d to ε is moderate: $\partial B_d/\partial \varepsilon \approx -0.11 \text{ MeV}$ per unit ε near $\varepsilon = 1.94$.

6 D-State Admixture Prediction

The antibonding admixture fraction (Conjecture 5.1) is a prediction of the deuteron’s D-state probability:

$$P_D = p = \frac{(\varepsilon - 1)^2}{3(\varepsilon + 1)^2} = \frac{0.8836}{3 \times 8.6436} = 0.0341 = 3.4\%. \quad (11)$$

The experimental D-state probability is model-dependent, ranging from $P_D \approx 4\%$ (Paris potential) to $P_D \approx 7\%$ (Reid soft-core), with a central value around 5.7% [3]. The CPP prediction of 3.4% is below the central experimental value but within the model-dependent range.

Remark 6.1 (Physical interpretation). *In standard nuclear physics, P_D arises from the tensor force (one-pion exchange) mixing $L = 2$ into the S -wave ground state. In CPP, the same mixing arises from the non-spherical (prolate) cage shape, which creates a quadrupole component in the inter-baryon DP chain coupling. The mechanism is different; the mathematical consequence — antibonding mode admixture reducing binding — is the same.*

7 Physical Interpretation

7.1 What physical objects are involved

The deuteron consists of two distorted tetrahedral cages, each housing three quarks and one open vertex, bonded through their open vertices by DP chain insertion. The energy is stored in the three quark-quark DP chain oscillators that form the contact face.

7.2 What they are doing

The three contact oscillators collectively vibrate in the K_3 bonding mode: all three in phase, like a breathing mode of the triangular face. This is the nuclear-physics analogue of the “S-wave” component of the deuteron.

The prolate cage shape introduces a small (3.4%) admixture of the antibonding modes, in which the oscillators vibrate out of phase. This is the CPP analogue of the “D-state.” The out-of-phase component partially cancels the in-phase binding, reducing the net energy from 2.342 to 2.222 MeV.

7.3 Where the energy is

The binding energy 2.222 MeV is the energy released when the two open vertices form a bond, converting unbound nucleons into a deuteron. This energy is stored in the collective oscillation of the three contact DP chains. It is a tiny fraction (0.24%) of the total nucleon mass energy, consistent with the deuteron being a weakly bound system.

7.4 The derivation chain

$$m_e \xrightarrow{z/\varphi} M_0 \xrightarrow{1/\varphi} \epsilon \xrightarrow{\lambda_+=2} B_d^{(0)} \xrightarrow{\kappa_d(\varepsilon)} B_d. \quad (12)$$

Each arrow is either a geometric operation (multiplication by lattice constants) or a spectral result (eigenvalue of K_3). The only non-geometric input is the cage distortion ε , which is itself derived from the force balance in SS-2.

8 CPP-to-Conventional-Physics Mapping

CPP concept	Conventional analogue	Status
Tetrahedral cage	Constituent quark model	Geometric
Open vertex	Nuclear bonding site	Derived (SS-2)
Base-to-base contact	S -wave nuclear configuration	Layer A
K_3 bonding mode	3S_1 partial wave	Layer B
Antibonding admixture p	D-state probability P_D	Conjecture
Prolate distortion ε	Tensor force (OPE)	Derived (SS-2)
M_0/φ	Deuteron binding scale	Derived
$\kappa_d = 0.949$	D-state correction	Conjecture
DP chain oscillation	Nuclear potential	Physical model

9 Scope and Open Problems

Claim	Status	Layer / Notes
K_3 contact face	Derived	Layer A (geometry)
$B_d^{(0)} = M_0/\varphi$	Proved	Layer C (Proposition 4.2)
$\kappa_d = 1 - (\varepsilon - 1)^2/[2(\varepsilon + 1)^2]$	Conjecture	Layer B (PT coefficient; Conjecture 5.1)
$B_d = 2.222$ MeV (-0.09%)	Conjecture	Layer C given Conjecture
$P_D = 3.4\%$	Conjecture	Identification with D-state (B3)
Deuteron radius	Open	Not attempted
Deuteron magnetic moment	Open	Not attempted
Extension to $A = 3, 4$	Open	See §10

9.1 Problem status after this paper

- **OPEN-SS-10** $\rightarrow$ **CONJ-SS-10**: Leading-order nuclear binding derived; NLO correction conjectured with -0.09% agreement.
- **New: OPEN-SS-17** (Deuteron radius): Predict $r_d = 2.128$ fm from the open-vertex model.
- **New: OPEN-SS-18** (Deuteron magnetic moment): Predict $\mu_d = 0.8574 \mu_N$ from the cage spin model.
- **New: OPEN-SS-19** (Rigorous NLO coefficient): Derive the exact coefficient in $p = C \cdot \alpha^2/\Delta$ from the cage-cage coupling Hamiltonian, promoting Conjecture 5.1 to a theorem.

10 Sketch: Extension to Light Nuclei

The base-to-base bonding mechanism extends naturally to light nuclei with $A = 3, 4$. We record the cascade formula here as a conjecture for future work; a full derivation is deferred.

Conjecture 10.1 (Light-nuclei cascade formula). *For closed-cascade light nuclei ($A = 2, 3, 4$):*

$$\boxed{B(A, Z) = B_0 C_{np}(A, Z) - P_0 C_\ell(A, Z) - E_C(A, Z) + B_0 \Delta_4(A)} \quad (13)$$

where

$$B_0 = \frac{M_0}{\varphi} = 2.342 \text{ MeV} \quad (\text{bonding quantum}), \quad (14)$$

$$P_0 = \frac{M_0}{\varphi^3} = \frac{B_0}{\varphi^2} = 0.894 \text{ MeV} \quad (\text{Pauli penalty}), \quad (15)$$

$$C_{np}(A, Z) = (A - 1) Z (A - Z) \quad (\text{proton-neutron cascade bonds}), \quad (16)$$

$$C_\ell(A, Z) = \binom{Z}{2} + \binom{A - Z}{2} \quad (\text{like-nucleon Pauli penalty}), \quad (17)$$

$$E_C(A, Z) = \binom{Z}{2} \frac{\alpha_{\text{em}} \hbar c}{1.2 A^{1/3} \text{ fm}} \quad (\text{Coulomb repulsion}), \quad (18)$$

$$\Delta_4(A) = \begin{cases} 1 & A = 4, \\ 0 & \text{otherwise.} \end{cases} \quad (\alpha\text{-particle shell closure}). \quad (19)$$

Remark 10.2 (Scope limitation). *This formula is intended only for the closed-cascade light-nucleus sector ($A = 2, 3, 4$), where the open-vertex graph is fully specified. Extension beyond $A = 4$ requires a separate polytope analysis.*

11 Conclusion

The deuteron binding energy has been derived from the K_3 spectral structure of the nuclear contact face in CPP:

$$B_d = \frac{m_e z}{\varphi^2} \left[1 - \frac{(\varepsilon - 1)^2}{2(\varepsilon + 1)^2} \right] = 2.222 \text{ MeV} \quad (\text{exp: } 2.2246 \text{ MeV, } -0.09\%). \quad (20)$$

The leading-order term M_0/φ comes from the K_3 bonding eigenvalue $\lambda_+ = 2$ acting on the pair-scale energy quantum. The next-to-leading-order correction comes from the cage distortion $\varepsilon = 1.94$ (SS-2), which induces a 3.4% antibonding admixture analogous to the deuteron D-state.

All inputs ($m_e, z, \varphi, \varepsilon$) are from the CPP framework. Zero parameters are specific to the deuteron. This is CPP's first prediction in nuclear physics — a new sector independent of the existing programme's lepton masses, gauge couplings, and quark mass hierarchy.

The NLO correction factor κ_d is registered as a conjecture: the perturbation-theory coefficient in $p = \alpha^2/\Delta$ is physically motivated but not rigorously derived from the cage-cage coupling Hamiltonian. A rigorous derivation (OPEN-SS-19) would promote the conjecture to a theorem.

The result adds a new star shot to the CPP programme's swarm validation: fifteen independent predictions from nine axioms now span particle masses, gauge couplings, lattice-scale structure, and nuclear binding.

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